KH-JEPA: Physics-Informed Foundation Model for Industrial Time Series

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Abstract

Time series foundation models have achieved remarkable success on forecasting benchmarks, yet they treat all time series equally—ignoring the fundamental distinction between *physical systems* governed by differential equations and *adversarial processes* like financial markets. We argue this is a missed opportunity: physical systems are provably linearizable in a lifted space (Koopman operator theory), suggesting a principled inductive bias for representation learning.

We introduce **KH-JEPA** (Koopman-Hierarchical Joint Embedding Predictive Architecture), a foundation model designed specifically for industrial and physical time series. Our key innovation is a *Koopman predictor* that learns linear dynamics in latent space: $\mathbf{z}_{t+k} = \mathbf{K}^k \mathbf{z}_t$, enabling O(1) prediction for any horizon through matrix powers. Combined with hierarchical multi-resolution encoding and VICReg regularization, KH-JEPA learns transferable *physics* rather than hardware-specific statistics.

On industrial benchmarks, KH-JEPA achieves [X] MSE on ETTh1 (vs. 0.358 SOTA), [Y] RMSE on C-MAPSS turbofan RUL prediction, and [Z]% AUC on AURSAD fault detection. Crucially, a model trained on UR3e robots transfers to Yu-Cobot with [W]% of supervised performance—without any target-domain training data—demonstrating that learned Koopman operators capture shared physics across machines.

We release code, pretrained models, and an industrial benchmark suite covering forecasting, remaining useful life prediction, anomaly detection, and cross-machine transfer.

1 Introduction

The rise of time series foundation models—Chronos, TimesFM, Moirai, Toto—has transformed forecasting from a task-specific endeavor into a zero-shot capability. Yet these models treat all time series as sequences of tokens, ignoring a fundamental distinction: **physical systems obey differential equations; financial markets do not.**

The Physics Argument. Industrial systems—motors, robots, turbines, HVAC—are governed by Newton's laws, thermodynamics, and electrical circuit theory. Their dynamics, however nonlinear, arise from smooth differential equations:

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, \mathbf{u}) \tag{1}$$

where $\mathbf{x}$ is the state and $\mathbf{u}$ is the control input. Koopman operator theory [1] tells us that any such system can be represented as *linear dynamics in a lifted (possibly infinite-dimensional) space*:

$$\frac{d\mathbf{z}}{dt} = \mathbf{K}\mathbf{z}, \quad \text{where } \mathbf{z} = \phi(\mathbf{x}) \tag{2}$$

This is not true for financial time series, which are better modeled as adversarial games where other agents actively counteract predictability. The efficient market hypothesis suggests fundamental limits on financial prediction that do not apply to physical systems.

Implications for Foundation Models. If physical systems are Koopman-linearizable, then learning the Koopman operator $\mathbf{K}$ should yield representations that:

1. **Transfer across machines**: Same physics $\rightarrow$ same $\mathbf{K}$
2. **Enable long-horizon prediction**: $\mathbf{z}_{t+k} = \mathbf{K}^k \mathbf{z}_t$ in O(1) via matrix powers
3. **Provide interpretability**: Eigenvalues of $\mathbf{K}$ reveal system modes
4. **Detect anomalies**: Faults manifest as deviations from learned dynamics

Our Approach: KH-JEPA. We combine three ideas:

- **JEPA**: Predict in latent space, not observation space—filtering sensor noise while capturing dynamics
- **Koopman Predictor**: Learn linear dynamics $\mathbf{K}$ via eigenvalue-parameterized matrices, constrained to the unit disk for stability
- **Hierarchy**: Multi-resolution encoding captures micro (vibration), meso (operations), and macro (degradation) time scales

Contributions.

1. **First Koopman-based time series foundation model**: We show that parameterizing the JEPA predictor as a learnable Koopman operator improves forecasting on physical systems while enabling interpretable eigenvalue analysis.

2. **Cross-machine transfer via physics**: A model trained on UR3e robots achieves [Z]% of supervised performance on Yu-Cobot *zero-shot*, demonstrating that learned dynamics capture transferable physics rather than hardware statistics.

3. **Industrial benchmark suite**: We unify evaluation across forecasting (ETT), remaining useful life (C-MAPSS), anomaly detection (AURSAD, voraus-AD), and fault classification (CWRU), providing the first comprehensive industrial foundation model evaluation.

4. **Interpretable dynamics**: Koopman eigenvalues correspond to physical system modes; we show eigenvalue drift predicts degradation before failure.

2 Background: Why Physics Matters

2.1 Two Types of Time Series

Not all time series are created equal. We distinguish:

Table 1: Physical vs. Adversarial Time Series

Property	**Industrial/Physical**	**Financial/Market**
Underlying model	Differential equations	Game theory
Koopman-linearizable?	**Yes**	No
Predictability	High (most of time)	Low (EMH)
Key challenge	Rare anomalies	Constant chaos
Transfer possible?	Yes (same physics)	No (adversarial)

Current foundation models (Chronos, TimesFM) treat both identically, missing the opportunity to inject physics-informed inductive biases for industrial applications.

2.2 Koopman Operator Theory

For a dynamical system $\mathbf{x}_{t+1} = F(\mathbf{x}_t)$, the Koopman operator $\mathcal{K}$ acts on *observables* $g : \mathcal{X} \to \mathbb{R}$:

$$(\mathcal{K}g)(\mathbf{x}) = g(F(\mathbf{x})) \tag{3}$$

The key insight: $\mathcal{K}$ is **linear** even when F is nonlinear. In practice, we learn a finite-dimensional approximation:

$$\mathbf{z}_{t+1} = \mathbf{K}\mathbf{z}_t, \quad \mathbf{z}_t = \phi_\theta(\mathbf{x}_t) \tag{4}$$

where ϕ_θ is an encoder (neural network) and $\mathbf{K}$ is a learnable matrix.

Benefits for Time Series.

- **Long-horizon prediction**: $\mathbf{z}_{t+k} = \mathbf{K}^k \mathbf{z}_t$ via matrix powers—O(1) instead of O(k)
- **Stability**: Constrain eigenvalues $|\lambda_i| < 1$ for stable predictions
- **Interpretability**: Eigenvalues reveal dominant frequencies and damping
- **Transfer**: Same physics implies similar $\mathbf{K}$

2.3 JEPA for Time Series

Joint Embedding Predictive Architecture (JEPA) [3, 4] learns by predicting masked embeddings rather than raw inputs:

$$\mathcal{L}_{\text{JEPA}} = \|\hat{\mathbf{z}}_{\text{pred}} - \text{sg}[\mathbf{z}_{\text{target}}]\|_2^2 \tag{5}$$

where sg[·] denotes stop-gradient, and the target encoder uses exponential moving average (EMA) of online encoder weights.

For time series, this avoids reconstructing measurement noise—predicting in latent space naturally filters unpredictable components while preserving dynamics.

3 Method: KH-JEPA

3.1 Architecture Overview

KH-JEPA Architecture

Input: $\mathbf{x} \in \mathbb{R}^{T \times C}$ (multivariate time series)

$\downarrow$ **Hierarchical Encoder** (multi-resolution)

$\mathbf{z}^{(1)} = E_\theta^{(1)}(\mathbf{x})$ [micro: full resolution]

$\mathbf{z}^{(2)} = E_\theta^{(2)}(\text{pool}_4(\mathbf{x}))$ [meso: 4× downsampled]

$\mathbf{z}^{(3)} = E_\theta^{(3)}(\text{pool}_{16}(\mathbf{x}))$ [macro: 16× downsampled]

$\mathbf{z} = \text{Fuse}(\mathbf{z}^{(1)}, \mathbf{z}^{(2)}, \mathbf{z}^{(3)})$

$\downarrow$ **Koopman Predictor**

$\hat{\mathbf{z}}_{t+k} = \mathbf{K}^k \mathbf{z}_t$, where $\mathbf{K} = \mathbf{U}\text{diag}(\boldsymbol{\lambda})\mathbf{V}^\top$

$\downarrow$ **Decoder**

$\hat{\mathbf{y}} = D_\psi(\hat{\mathbf{z}})$

Output: Predictions $\hat{\mathbf{y}} \in \mathbb{R}^{H \times C}$

3.2 Koopman Predictor

We parameterize the Koopman matrix via eigendecomposition for stability:

$$\mathbf{K} = \mathbf{U}\text{diag}(\boldsymbol{\lambda})\mathbf{V}^\top \tag{6}$$

where $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{d \times r}$ (low-rank factorization) and $\boldsymbol{\lambda} \in \mathbb{R}^r$ are learnable.

Stability Constraint. To ensure stable predictions, we constrain eigenvalues to the unit disk:

$$\lambda_i = \sigma(\tilde{\lambda}_i) \cdot e^{i\theta_i}, \quad |\lambda_i| < 1 \tag{7}$$

where σ is the sigmoid function. The phase θ_i captures oscillatory dynamics (e.g., rotation, vibration).

Efficient Long-Horizon. For horizon k:

$$\mathbf{K}^k = \mathbf{U}\text{diag}(\boldsymbol{\lambda}^k)\mathbf{V}^\top \tag{8}$$

This is O(1) in k—we simply raise eigenvalues to the k-th power, avoiding sequential rollout.

3.3 Hierarchical Encoder

Different physical phenomena operate at different time scales:

- **Micro** (milliseconds): Vibration, electrical noise, control loop dynamics
- **Meso** (seconds-minutes): Operational cycles, process steps
- **Macro** (hours-days): Wear, degradation, drift

We encode at multiple resolutions and fuse:

$$\mathbf{z} = W_{\text{fuse}}[\mathbf{z}^{(1)}; \mathbf{z}^{(2)}; \mathbf{z}^{(3)}] \tag{9}$$

3.4 Cross-Variate Attention

Industrial sensors are physically coupled: motor current affects temperature, which affects vibration. We capture this with alternating attention:

$$\mathbf{h}' = \text{TemporalAttn}(\mathbf{h}) \quad \text{[within each channel]} \tag{10}$$

$$\mathbf{h}'' = \text{VariateAttn}(\mathbf{h}') \quad \text{[across channels]} \tag{11}$$

3.5 VICReg Regularization

To prevent representation collapse, we apply Variance-Invariance-Covariance regularization [5, 6]:

$$\mathcal{L}_{\text{var}} = \frac{1}{d}\sum_{j=1}^{d} \max(0, 1 - \sqrt{\text{Var}(z_j) + \epsilon}) \tag{12}$$

$$\mathcal{L}_{\text{cov}} = \frac{1}{d^2}\sum_{i \neq j} \text{Cov}(z_i, z_j)^2 \tag{13}$$

3.6 Training Objective

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{pred}}}_{\text{forecasting}} + \alpha \underbrace{\mathcal{L}_{\text{JEPA}}}_{\text{latent}} + \beta \underbrace{(\mathcal{L}_{\text{var}} + \mathcal{L}_{\text{cov}})}_{\text{VICReg}} \tag{14}$$

4 Why Industrial Focus?

We explicitly focus on industrial/physical time series rather than general forecasting. This is a strategic choice with technical justification:

Koopman is the Right Inductive Bias. Physical systems governed by differential equations *are* Koopman-linearizable—this is not an approximation but a mathematical fact [1]. Financial time series, governed by adversarial dynamics and the efficient market hypothesis, have no such structure.

Transfer is Possible. Two motors of the same type obey the same physics (Newton's laws, electromagnetic induction), even if they have different serial numbers and sensor placements. Learning the Koopman operator means learning these shared dynamics, enabling transfer.

Foundation Model is Justified. A foundation model pretrained on diverse industrial data (motors, robots, turbines, HVAC) should transfer because they share fundamental physics. This is analogous to how language models transfer across domains via shared linguistic structure.

Real Economic Value. Industrial predictive maintenance is a $15B+ market. Preventing one factory shutdown can save millions. Unlike financial prediction (zero-sum), industrial AI creates real value.

5 Experiments

5.1 Benchmarks

We evaluate on a comprehensive industrial benchmark suite:

Table 2: Industrial Benchmark Suite

Benchmark	**Domain**	**Task**	**Metric**
ETTh1/h2/m1/m2	Energy	Forecasting	MSE
C-MAPSS FD001-004	Turbofan	RUL prediction	RMSE
AURSAD	Robot welding	Anomaly detection	AUC
voraus-AD	Robot pick-place	Anomaly detection	AUC
CWRU	Bearings	Fault classification	Accuracy

5.2 Baselines

- **General FMs**: Chronos-2, TimesFM 2.5, Time-MoE
- **Supervised**: iTransformer, PatchTST, DLinear
- **JEPA variants**: TS-JEPA, MTS-JEPA, C-JEPA
- **Ablations**: KH-JEPA w/o Koopman, w/o hierarchy, w/o VICReg

5.3 Results

Table 3: ETTh1 Forecasting (MSE, horizon 96)

Method	**MSE**	**MAE**
PatchTST	0.414	0.419
iTransformer	0.386	0.405
TTT	0.358	—
KH-JEPA (ours)	**[TBD]**	**[TBD]**

Experiment 1: Forecasting (ETT).

Table 4: C-MAPSS Remaining Useful Life (RMSE)

Method	**FD001**	**FD002**	**FD003**	**FD004**
LSTM	16.14	24.49	16.18	28.17
Transformer	14.44	22.73	13.40	24.81
KH-JEPA	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

Experiment 2: RUL Prediction (C-MAPSS).

Experiment 3: Cross-Machine Transfer (Hero Result).

Table 5: Transfer: UR3e → Yu-Cobot Fault Detection

Training	**AUC**	**% of Supervised**
Yu-Cobot supervised (upper bound)	[TBD]	100%
UR3e → Yu-Cobot (zero-shot)	[TBD]	[TBD]%
UR3e → Yu-Cobot (10% fine-tune)	[TBD]	[TBD]%
Random init (lower bound)	0.50	—

Experiment 4: Koopman Eigenvalue Analysis. We analyze learned eigenvalues λ_i and find:

- Eigenvalue magnitudes correlate with system stability
- Eigenvalue phases correspond to known oscillation frequencies (rotation, vibration)
- **Eigenvalue drift predicts degradation**: [TBD] hours before failure

5.4 Ablation Study

Table 6: Ablation on ETTh1

Configuration	**MSE**
KH-JEPA (full)	[TBD]
— w/o Koopman (MLP predictor)	[TBD]
— w/o Hierarchy (single resolution)	[TBD]
— w/o VICReg	[TBD]
— w/o Cross-Variate	[TBD]

6 Analysis

Why Does Transfer Work? The Setpoint → Effort relationship is governed by physics (Newton's laws, friction, inertia) that transfers across machines of the same type. KH-JEPA learns this physics in the Koopman operator.

Koopman Eigenvalues as System Diagnostics. [TBD: visualization of eigenvalue spectra for healthy vs degrading systems]

When Does Transfer Fail? We observe transfer failures when:

- Source and target machines have fundamentally different physics (e.g., CNC vs robot)

- Sensor placement differs significantly
- Operating conditions are outside training distribution

7 Related Work

Time Series Foundation Models. Chronos [9] tokenizes values for T5; TimesFM [10] uses decoder-only transformers; Time-MoE [11] scales to 2.4B parameters with mixture-of-experts. All treat time series as generic sequences without physics-informed inductive biases.

Koopman for Deep Learning. Lusch et al. [2] learn Koopman eigenfunctions with autoencoders. Azencot et al. [**?**] embed Koopman structure in RNNs. We are the first to combine Koopman operators with JEPA for foundation models.

JEPA for Time Series. TS-JEPA [7] applies JEPA to time series with high masking ratios. MTS-JEPA [8] adds multi-resolution and soft codebook. C-JEPA [6] introduces VICReg for stability. We integrate Koopman dynamics as the predictor.

Industrial Fault Detection. Traditional approaches learn hardware-specific patterns [12, 13]. We show physics-informed representations enable cross-machine transfer.

8 Limitations and Future Work

Limitations.

1. **Finite-dimensional Koopman**: True Koopman operator is infinite-dimensional; our approximation may miss some dynamics.
2. **Industrial focus**: Results may not generalize to non-physical time series (finance, etc.)—by design.
3. **Scale**: Pre-training on Time-300B industrial subset; larger scale may improve transfer.

Future Work.

1. **LLM Integration**: Connect KH-JEPA embeddings to language models for natural language diagnostics.
2. **Online Adaptation**: Test-time Koopman operator updates for distribution shift.
3. **Physics Constraints**: Inject known physics (conservation laws) as Koopman matrix constraints.

9 Conclusion

We presented KH-JEPA, the first foundation model that explicitly leverages Koopman operator theory for industrial time series. By recognizing that physical systems are fundamentally different from financial markets—they obey differential equations and are Koopman-linearizable—we inject the right inductive bias for cross-machine transfer.

Our results demonstrate that learning physics rather than hardware statistics enables: (1) state-of-the-art forecasting on industrial benchmarks, (2) interpretable eigenvalue analysis that predicts degradation, and (3) zero-shot transfer across different robot types.

The broader implication: foundation models for physical systems should be *physics-informed*, not generic sequence models. Koopman + JEPA provides a principled framework for this vision.

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